Lab 3: Component Modelling & Architectural Pattern Selection

Deliverable 1: UML 2.0 Component Diagram (Self-Service Coffee Kiosk System)

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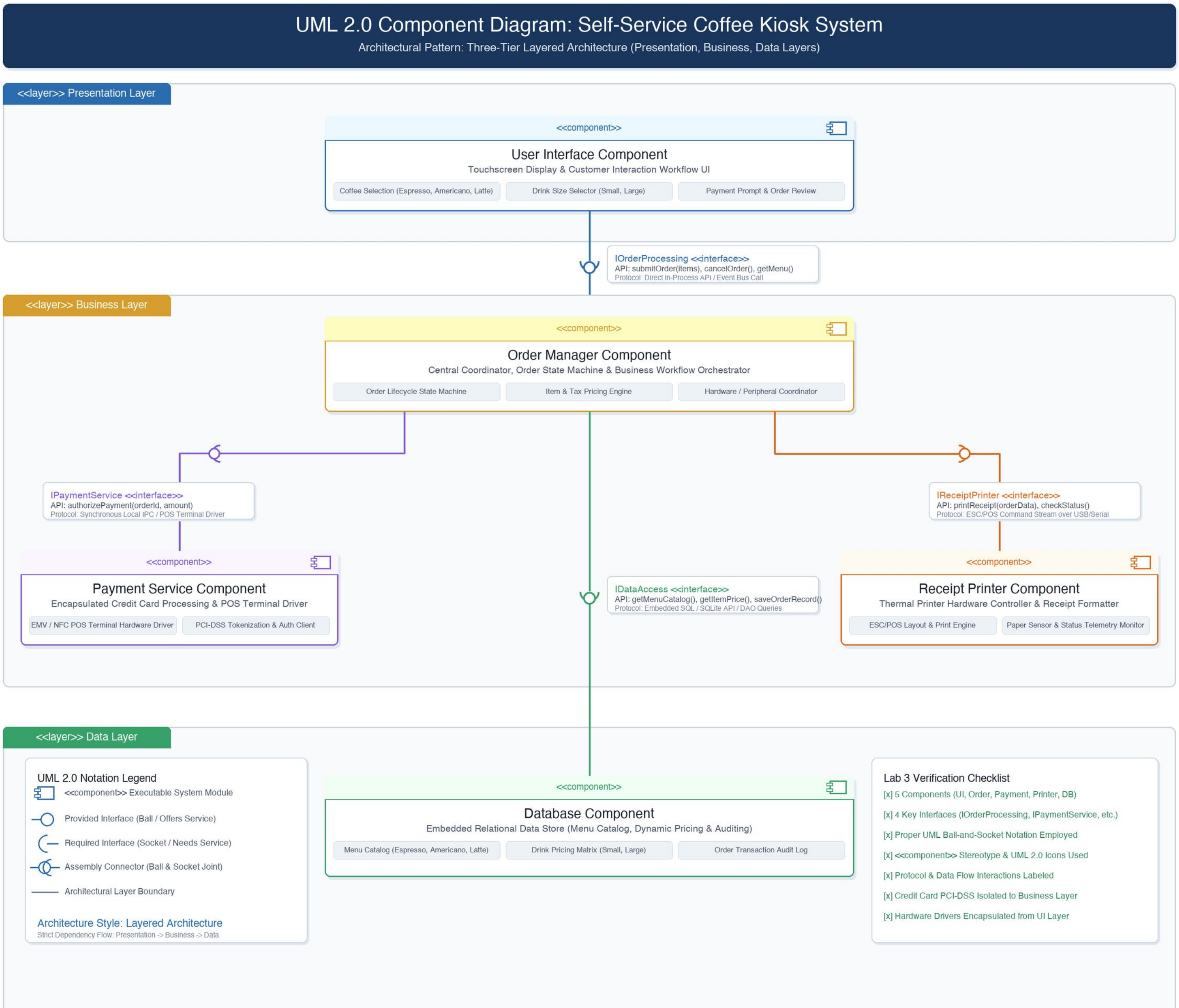

Deliverable 1 Technical Specifications & Traceability

- Architectural Style: Three-Tier Layered Architecture (Presentation Layer, Business Layer, Data Layer).
- 5 Components Modeled: User Interface Component, Order Manager (given), Payment Service (given), Receipt Printer, Database.
- 4 Standard Interfaces: IOrderProcessing (API), IPaymentService (POS Terminal IPC), IReceiptPrinter (ESC/POS USB), IDataAccess (SQLite DAO).
- UML 2.0 Compliance: Official ball-and-socket notation, <<component>> stereotypes, UML 2.0 component glyphs, and layered packages.

Lab 3: Architecture Selection & Component Modelling

Architecture Selection: "We chose Layered Architecture for the Self-Service Coffee Kiosk System."

System: Self-Service Coffee Kiosk | Target: Touchscreen POS & Printer | Pattern: 3-Tier Layered

1. ARCHITECTURAL CHOICE

We have selected a Three-Tier Layered Architecture structured into horizontal tiers: Presentation Layer (Touchscreen UI), Business Layer (Order Manager, Payment Service, Receipt Printer), and Data Layer (Database Component). Each layer enforces strict separation of concerns, communicating unidirectionally through standardized, decoupled ball-and-socket interfaces.

2. TWO SPECIFIC SCENARIO-RELATED REASONS FOR SELECTION

- Reason 1: Hardware Abstraction & Peripheral Isolation (Touchscreen & Thermal Printer): The coffee kiosk physically integrates specialized hardware devices (an interactive touchscreen and an ESC/POS receipt printer). A layered pattern cleanly isolates low-level hardware dependencies within the Business/Infrastructure layer. The ReceiptPrinter component encapsulates all device escape codes, USB/Serial polling, and paper-out sensor telemetry. Consequently, the touchscreen UI never interacts directly with hardware drivers, preventing UI freezing during mechanical print jobs and allowing printer hardware replacement without affecting the ordering interface.
- Reason 2: Domain Modularity for Menu Configurations & Dynamic Pricing: The scenario demands handling specific drink combinations (Espresso, Americano, Latte across Small and Large sizes) with dynamic pricing and credit-card-only checkout. Layered architecture cleanly decouples the presentation screens from order validation and cost calculations. The OrderManager coordinates product sizing and total computation by querying the Database component, ensuring that menu changes, price adjustments, or seasonal drink additions require zero modifications to the touchscreen presentation codebase.

3. SECURITY ADVANTAGE (CREDIT CARD & PCI-DSS ISOLATION)

Because customer payment is restricted to credit card only, PCI-DSS compliance and cardholder data security are critical. In this layered architecture, credit card handling is completely isolated inside the PaymentService component behind a strictly typed IPaymentService interface connected directly to the POS EMV/NFC card reader. Neither the UserInterface component nor the Database component has access to raw card numbers or CVV codes. Sensitive cryptographic tokens never cross layer boundaries, preventing memory scraping attacks from UI widgets and preventing accidental storage of sensitive payment tokens in local audit tables.

4. PERFORMANCE BENEFIT (TOUCHSCREEN RESPONSIVENESS & IN-MEMORY CACHING)

In a busy cafe, high customer ordering throughput requires near-instantaneous touchscreen response ($<$ 100ms latency). Operating as a localized layered system allows the OrderManager to cache static menu and pricing tables in local RAM upon boot, completely bypassing repeated disk queries while customers browse. Furthermore, layer boundaries enable asynchronous, non-blocking I/O: time-consuming operations (POS card authorization and mechanical receipt printing) are dispatched to background worker threads while the UI immediately transitions to animated customer feedback screens without dropping display frames.

5. ARCHITECTURAL SUMMARY MAPPING

Layer	Component Name	Key System Responsibility	Interfaces / Protocols
Presentation	User Interface Component	Touchscreen customer ordering UI (3 coffees, 2 sizes)	Requires: IOrderProcessing
Business	Order Manager Component	Order workflow orchestration, pricing, and state machine	Provides: IOrderProcessing | Requires: 3 interfaces
Business	Payment Service Component	Credit card POS processing & PCI-DSS compliance	Provides: IPaymentService (Local Encrypted IPC)
Business	Receipt Printer Component	Thermal printer driver & ESC/POS layout formatting	Provides: IReceiptPrinter (USB/Serial Bus)
Data	Database Component	Menu catalog, drink pricing matrix & transaction audit log	Provides: IDataAccess (SQLite / DAO Queries)